

Algebra 1 — Full Practice Workbook

All 9 chapters · ~50 problems · Step-by-step worked solutions · Neil Velicheti · April 2026

How to use this workbook. Work each chapter in order. Print the chapter, work the problems in pen on the lines provided, and check yourself against the Solutions section at the back. Don't peek before trying. This complements the dedicated Chapter 8 Workbook — for deeper Ch 8 practice, use that one.

Format. Each chapter starts with a brief reminder of the rules. Problems are graduated from foundational to harder/applied. All solutions are at the end so the front of the book can be used as a clean practice test.

Pacing recommendation. One chapter per day during summer (Phase 2). About 30–40 minutes per chapter. Score yourself: Mastered $\geq 90\%$, Strong 80–89%, Okay 70–79%, Needs review $< 70\%$. Mark each chapter on the Mastery tab of the Tracker spreadsheet.

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Chapter 1 — Solving Linear Equations

Goal: isolate the variable. Use inverse operations in reverse PEMDAS order. Distribute first if needed; combine like terms; move variables to one side; solve.

Problem 1.1

Solve: $3(x - 4) + 2x = 17$

Problem 1.2

Solve: $4x - 7 = 2x + 9$

Problem 1.3

Solve: $(x/3) + 5 = 11$

Problem 1.4

Solve the literal equation for y: $3x + 4y = 12$

Problem 1.5

Solve: $2(3x - 5) - 4(x + 1) = 6$

Problem 1.6

Word problem: A taxi charges a \$3 flat fee plus \$2 per mile. If your total ride was \$19, how many miles did you travel?

Chapter 2 — Inequalities

Solve like equations BUT flip the inequality sign whenever you multiply or divide both sides by a negative number. For compound 'and' (between), solve all parts together; for 'or', solutions can satisfy either.

Problem 2.1

Solve and graph the solution: $-3x + 5 \leq 14$

Problem 2.2

Solve: $2(x - 3) > x + 4$

Problem 2.3

Solve the compound inequality: $-1 \leq 2x + 3 < 9$

Problem 2.4

Solve the compound inequality (or): $x - 4 < -2$ OR $3x \geq 15$

Problem 2.5

Solve the absolute-value inequality: $|2x - 1| < 7$

Chapter 3 — Functions & Function Notation

A relation is a function if every input has exactly one output (vertical line test on a graph). Use $f(x)$ notation: $f(3)$ means 'plug 3 in for x .' Domain = set of allowed inputs; range = set of outputs.

Problem 3.1

If $f(x) = 2x^2 - 5x + 1$, find $f(3)$ and $f(-2)$.

Problem 3.2

If $f(x) = 3x - 7$ and $g(x) = x^2 + 2$, find $f(g(2))$.

Problem 3.3

Determine whether $\{(1, 2), (3, 5), (1, 7), (4, 9)\}$ is a function. Explain.

Problem 3.4

If $f(x) = 2x + 5$, find $f(x + 3)$ in simplified form.

Problem 3.5

State the domain and range of the function shown by the points: $(-2, 5)$, $(0, 1)$, $(3, -4)$, $(4, 7)$.

Chapter 4 — Linear Functions & Graphing

Slope $m = (y_2 - y_1)/(x_2 - x_1)$. Slope-intercept form $y = mx + b$. Point-slope form $y - y_1 = m(x - x_1)$. Parallel lines have equal slopes; perpendicular slopes multiply to -1 .

Problem 4.1

Find the slope between $(-2, 5)$ and $(4, -7)$.

Problem 4.2

Write the equation of the line through $(3, -1)$ with slope 2 in slope-intercept form.

Problem 4.3

Write the equation of the line through $(1, 4)$ and $(5, -4)$ in slope-intercept form.

Problem 4.4

Write the equation of the line parallel to $y = (1/3)x - 2$ that passes through $(6, 5)$.

Problem 4.5

Write the equation of the line perpendicular to $y = (1/2)x + 4$ passing through $(-2, 3)$.

Problem 4.6

Convert $4x - 3y = 12$ to slope-intercept form. State slope and y-intercept.

Chapter 5 — Systems of Equations

A system of equations has TWO equations with TWO unknowns. Substitution: solve one for a variable and plug in. Elimination: add or subtract the equations (after scaling) to cancel a variable. One solution = lines intersect; no solution = parallel; infinitely many = same line.

Problem 5.1

Solve by substitution: $y = 2x + 1$ and $3x + y = 16$

Problem 5.2

Solve by elimination: $2x + 3y = 12$ and $2x - y = 4$

Problem 5.3

Solve: $3x + 2y = 11$ and $5x - 4y = 11$

Problem 5.4

Word problem: A coffee shop sells small drinks for \$3 and large drinks for \$5. On Monday they sold 80 drinks for \$310. How many of each were sold?

Problem 5.5

How many solutions? $2x - 3y = 6$ and $-4x + 6y = -12$

Chapter 6 — Exponents & Exponential Functions

Exponent rules: same base $\rightarrow$ add exponents (multiplication) or subtract (division). Power of a power: multiply exponents. Negative exponents flip; zero exponent = 1. Exponential growth: $y = a(1 + r)^t$. Decay: $y = a(1 - r)^t$.

Problem 6.1

Simplify: $(3x^2y^3)(4x^5y)$

Problem 6.2

Simplify: $(2x^3)^4$

Problem 6.3

Simplify: $(x^5y^2) / (x^2y^{-1})$

Problem 6.4

Write 0.00045 in scientific notation.

Problem 6.5

Exponential decay: A car's value depreciates 12% per year. It originally costs \$25,000. Write the function and find its value after 5 years.

Problem 6.6

Find the next term in the geometric sequence: 3, 12, 48, 192, ...

Chapter 7 — Polynomials & Factoring

Always factor out the GCF first. Patterns: difference of squares $a^2 - b^2 = (a+b)(a-b)$; perfect square trinomial $a^2 \pm 2ab + b^2 = (a \pm b)^2$; trinomial $x^2 + bx + c \rightarrow$ find two numbers product c , sum b . For $ax^2 + bx + c$ with $a > 1$, use AC method.

Problem 7.1

Multiply: $(2x - 3)(x + 5)$

Problem 7.2

Add: $(3x^2 - 4x + 7) + (x^2 + 2x - 5)$

Problem 7.3

Factor: $x^2 + 8x + 15$

Problem 7.4

Factor: $x^2 - 5x - 24$

Problem 7.5Factor: $6x^2 + 11x - 10$

Problem 7.6Factor: $9x^2 - 49$

Problem 7.7Factor completely: $3x^3 - 12x$

Problem 7.8Factor: $x^2 + 14x + 49$

Chapter 8 — Quadratic Functions & Equations

Standard form $y = ax^2 + bx + c$. Vertex form $y = a(x - h)^2 + k$. Axis of symmetry $x = -b/(2a)$. Discriminant $D = b^2 - 4ac$. To solve quadratics: factor, square roots, complete the square, or quadratic formula. For deeper Ch 8 practice see the dedicated Ch 8 Workbook.

Problem 8.1

Identify the vertex and axis of symmetry of $y = (x - 4)^2 + 7$.

Problem 8.2

Solve by factoring: $x^2 - 7x + 12 = 0$

Problem 8.3

Solve by quadratic formula: $3x^2 + 5x - 2 = 0$

Problem 8.4

Find the discriminant of $y = 4x^2 - 12x + 9$ and number of real solutions.

Problem 8.5

A rocket is launched from a 6-foot-high platform at 64 ft/s. Its height is $h(t) = -16t^2 + 64t + 6$. What is the maximum height?

Problem 8.6

Convert to vertex form by completing the square: $y = x^2 - 6x + 11$

Chapter 9 — Radicals, Pythagorean & Distance

Simplify radicals by factoring out perfect squares. Rationalize denominators by multiplying by the radical.
Pythagorean: $a^2 + b^2 = c^2$ (right triangles). Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Problem 9.1

Simplify: $\sqrt{72}$

Problem 9.2

Simplify: $\sqrt{20} + \sqrt{45}$

Problem 9.3

Rationalize: $4 / \sqrt{3}$

Problem 9.4

A right triangle has legs 8 and 15. Find the hypotenuse.

Problem 9.5

Find the distance between (1, -3) and (5, 9).

Problem 9.6

Find the midpoint of $(-4, 7)$ and $(10, -1)$.

Solutions (Worked, Step-by-Step)

Don't read until you've tried the problem yourself. Each step shows reasoning, not just math.

Chapter 1 — Solving Linear Equations

Problem 1.1

Solve: $3(x - 4) + 2x = 17$

Distribute: $3x - 12 + 2x = 17$.

Combine: $5x - 12 = 17$.

Add 12: $5x = 29$.

Divide by 5: $x = 29/5$ (or 5.8).

Answer: $x = 29/5$

Problem 1.2

Solve: $4x - 7 = 2x + 9$

Subtract $2x$: $2x - 7 = 9$.

Add 7: $2x = 16$.

Divide: $x = 8$.

Answer: $x = 8$

Problem 1.3

Solve: $(x/3) + 5 = 11$

Subtract 5: $x/3 = 6$.

Multiply by 3: $x = 18$.

Answer: $x = 18$

Problem 1.4

Solve the literal equation for y : $3x + 4y = 12$

Subtract $3x$: $4y = 12 - 3x$.

Divide by 4: $y = (12 - 3x)/4 = 3 - (3/4)x$.

Answer: $y = 3 - (3/4)x$

Problem 1.5

Solve: $2(3x - 5) - 4(x + 1) = 6$

Distribute: $6x - 10 - 4x - 4 = 6$.

Combine: $2x - 14 = 6$.

Add 14: $2x = 20$.

Divide: $x = 10$.

Answer: $x = 10$

Problem 1.6

Word problem: A taxi charges a \$3 flat fee plus \$2 per mile. If your total ride was \$19, how many miles did you travel?

Let m = miles. Equation: $3 + 2m = 19$.

Subtract 3: $2m = 16$.

Divide: $m = 8$.

Answer: 8 miles

Chapter 2 — Inequalities

Problem 2.1

Solve and graph the solution: $-3x + 5 \leq 14$

Subtract 5: $-3x \leq 9$.

Divide by -3 (flip!): $x \geq -3$.

Answer: $x \geq -3$ (closed dot at -3 , arrow right)

Problem 2.2

Solve: $2(x - 3) > x + 4$

Distribute: $2x - 6 > x + 4$.

Subtract x : $x - 6 > 4$.

Add 6: $x > 10$.

Answer: $x > 10$

Problem 2.3

Solve the compound inequality: $-1 \leq 2x + 3 < 9$

Subtract 3 from each part: $-4 \leq 2x < 6$.

Divide each by 2: $-2 \leq x < 3$.

Answer: $-2 \leq x < 3$

Problem 2.4

Solve the compound inequality (or): $x - 4 < -2$ OR $3x \geq 15$

Left: $x - 4 < -2 \rightarrow x < 2$.

Right: $3x \geq 15 \rightarrow x \geq 5$.

Answer: $x < 2$ OR $x \geq 5$

Problem 2.5

Solve the absolute-value inequality: $|2x - 1| < 7$

Rewrite: $-7 < 2x - 1 < 7$.

Add 1 to each: $-6 < 2x < 8$.

Divide by 2: $-3 < x < 4$.

Answer: $-3 < x < 4$

Chapter 3 — Functions & Function Notation

Problem 3.1

If $f(x) = 2x^2 - 5x + 1$, find $f(3)$ and $f(-2)$.

$$f(3) = 2(9) - 5(3) + 1 = 18 - 15 + 1 = 4.$$

$$f(-2) = 2(4) - 5(-2) + 1 = 8 + 10 + 1 = 19.$$

Answer: $f(3) = 4$; $f(-2) = 19$

Problem 3.2

If $f(x) = 3x - 7$ and $g(x) = x^2 + 2$, find $f(g(2))$.

$$\text{First } g(2) = 4 + 2 = 6.$$

$$\text{Then } f(6) = 3(6) - 7 = 11.$$

Answer: $f(g(2)) = 11$

Problem 3.3

Determine whether $\{(1, 2), (3, 5), (1, 7), (4, 9)\}$ is a function. Explain.

$x = 1$ maps to two different y -values (2 and 7). Not a function.

Answer: Not a function: input 1 has two outputs.

Problem 3.4

If $f(x) = 2x + 5$, find $f(x + 3)$ in simplified form.

$$\text{Replace } x \text{ with } (x + 3): f(x + 3) = 2(x + 3) + 5 = 2x + 6 + 5 = 2x + 11.$$

Answer: $f(x + 3) = 2x + 11$

Problem 3.5

State the domain and range of the function shown by the points: $(-2, 5), (0, 1), (3, -4), (4, 7)$.

Domain = set of x -values: $\{-2, 0, 3, 4\}$.

Range = set of y -values: $\{-4, 1, 5, 7\}$.

Answer: Domain $\{-2, 0, 3, 4\}$; Range $\{-4, 1, 5, 7\}$.

Chapter 4 — Linear Functions & Graphing

Problem 4.1

Find the slope between $(-2, 5)$ and $(4, -7)$.

$$m = (y_2 - y_1)/(x_2 - x_1) = (-7 - 5)/(4 - (-2)) = -12/6.$$

Answer: $m = -2$

Problem 4.2

Write the equation of the line through $(3, -1)$ with slope 2 in slope-intercept form.

$$y = mx + b \rightarrow -1 = 2(3) + b \rightarrow -1 = 6 + b \rightarrow b = -7.$$

Answer: $y = 2x - 7$

Problem 4.3

Write the equation of the line through (1, 4) and (5, -4) in slope-intercept form.

$$\text{Slope: } m = (-4 - 4)/(5 - 1) = -8/4 = -2.$$

$$\text{Use (1, 4): } 4 = -2(1) + b \rightarrow b = 6.$$

Answer: $y = -2x + 6$

Problem 4.4

Write the equation of the line parallel to $y = (1/3)x - 2$ that passes through (6, 5).

$$\text{Parallel} \rightarrow \text{same slope, } m = 1/3.$$

$$\text{Use point (6, 5): } 5 = (1/3)(6) + b \rightarrow 5 = 2 + b \rightarrow b = 3.$$

Answer: $y = (1/3)x + 3$

Problem 4.5

Write the equation of the line perpendicular to $y = (1/2)x + 4$ passing through (-2, 3).

$$\text{Perpendicular slope is the negative reciprocal: } m = -2.$$

$$\text{Use point (-2, 3): } 3 = -2(-2) + b \rightarrow 3 = 4 + b \rightarrow b = -1.$$

Answer: $y = -2x - 1$

Problem 4.6

Convert $4x - 3y = 12$ to slope-intercept form. State slope and y-intercept.

$$\text{Subtract } 4x: -3y = -4x + 12.$$

$$\text{Divide by } -3: y = (4/3)x - 4.$$

Answer: Slope 4/3; y-intercept -4.

Chapter 5 — Systems of Equations

Problem 5.1

Solve by substitution: $y = 2x + 1$ and $3x + y = 16$

$$\text{Substitute first into second: } 3x + (2x + 1) = 16.$$

$$5x + 1 = 16 \rightarrow 5x = 15 \rightarrow x = 3.$$

$$y = 2(3) + 1 = 7.$$

Answer: (3, 7)

Problem 5.2

Solve by elimination: $2x + 3y = 12$ and $2x - y = 4$

$$\text{Subtract second from first: } (2x + 3y) - (2x - y) = 12 - 4.$$

$$4y = 8 \rightarrow y = 2.$$

$$\text{Substitute: } 2x - 2 = 4 \rightarrow x = 3.$$

Answer: (3, 2)

Problem 5.3

Solve: $3x + 2y = 11$ and $5x - 4y = 11$

Multiply first by 2: $6x + 4y = 22$.

Add to second: $11x = 33 \rightarrow x = 3$.

Substitute: $3(3) + 2y = 11 \rightarrow 9 + 2y = 11 \rightarrow y = 1$.

Answer: (3, 1)

Problem 5.4

Word problem: A coffee shop sells small drinks for \$3 and large drinks for \$5. On Monday they sold 80 drinks for \$310. How many of each were sold?

Let s = small, L = large. $s + L = 80$; $3s + 5L = 310$.

From first: $s = 80 - L$. Substitute: $3(80 - L) + 5L = 310$.

$240 - 3L + 5L = 310 \rightarrow 2L = 70 \rightarrow L = 35$.

$s = 80 - 35 = 45$.

Answer: 45 small, 35 large

Problem 5.5

How many solutions? $2x - 3y = 6$ and $-4x + 6y = -12$

Multiply first by -2 : $-4x + 6y = -12$. Same as second.

The two equations are equivalent $\rightarrow$ same line.

Answer: Infinitely many solutions

Chapter 6 — Exponents & Exponential Functions

Problem 6.1

Simplify: $(3x^2y^3)(4x^5y)$

Multiply coefficients: $3 \cdot 4 = 12$.

Add exponents on like bases: $x^2 \cdot x^5 = x^7$; $y^3 \cdot y^1 = y^4$.

Answer: $12x^7y^4$

Problem 6.2

Simplify: $(2x^3)^4$

Power of a product: $2^4 \cdot (x^3)^4$.

$= 16 \cdot x^{12} = 16x^{12}$.

Answer: $16x^{12}$

Problem 6.3

Simplify: $(x^5y^2) / (x^2y^{-1})$

Subtract exponents: $x^{(5-2)} = x^3$; $y^{(2-(-1))} = y^3$.

Answer: x^3y^3

Problem 6.4

Write 0.00045 in scientific notation.

Move decimal 4 places to the right: 4.5. The exponent is -4 .

Answer: 4.5×10^{-4}

Problem 6.5

Exponential decay: A car's value depreciates 12% per year. It originally costs \$25,000. Write the function and find its value after 5 years.

Decay: $y = 25000(1 - 0.12)^t = 25000(0.88)^t$.

At $t = 5$: $25000(0.88)^5 \approx 25000(0.5277) \approx 13,193$.

Answer: $y = 25000(0.88)^t$; $\approx \$13,193$ after 5 years.

Problem 6.6

Find the next term in the geometric sequence: 3, 12, 48, 192, ...

Common ratio $r = 12/3 = 4$.

Next term: $192 \cdot 4 = 768$.

Answer: 768

Chapter 7 — Polynomials & Factoring

Problem 7.1

Multiply: $(2x - 3)(x + 5)$

FOIL: $2x \cdot x + 2x \cdot 5 - 3 \cdot x - 3 \cdot 5 = 2x^2 + 10x - 3x - 15$.

Combine: $2x^2 + 7x - 15$.

Answer: $2x^2 + 7x - 15$

Problem 7.2

Add: $(3x^2 - 4x + 7) + (x^2 + 2x - 5)$

Combine like terms: $(3+1)x^2 + (-4+2)x + (7-5)$.

Answer: $4x^2 - 2x + 2$

Problem 7.3

Factor: $x^2 + 8x + 15$

Need two numbers with product 15, sum 8 $\rightarrow$ 3 and 5.

Answer: $(x + 3)(x + 5)$

Problem 7.4

Factor: $x^2 - 5x - 24$

Two numbers with product -24 , sum $-5 \rightarrow -8$ and 3.

Answer: $(x - 8)(x + 3)$

Problem 7.5Factor: $6x^2 + 11x - 10$ AC = $6 \cdot (-10) = -60$. Need numbers with product -60 , sum $11 \rightarrow 15$ and -4 .Split middle: $6x^2 + 15x - 4x - 10$.Group: $3x(2x + 5) - 2(2x + 5) = (3x - 2)(2x + 5)$.**Answer: $(3x - 2)(2x + 5)$** **Problem 7.6**Factor: $9x^2 - 49$ Difference of squares: $(3x)^2 - (7)^2$.**Answer: $(3x + 7)(3x - 7)$** **Problem 7.7**Factor completely: $3x^3 - 12x$ GCF $3x$: $3x(x^2 - 4)$. $x^2 - 4$ is a difference of squares: $(x + 2)(x - 2)$.**Answer: $3x(x + 2)(x - 2)$** **Problem 7.8**Factor: $x^2 + 14x + 49$ Perfect square trinomial: $a = x$, $b = 7$ (since $2 \cdot 7 = 14$ and $7^2 = 49$).**Answer: $(x + 7)^2$** **Chapter 8 — Quadratic Functions & Equations****Problem 8.1**Identify the vertex and axis of symmetry of $y = (x - 4)^2 + 7$.Vertex form $y = a(x - h)^2 + k \rightarrow$ vertex $(4, 7)$; axis $x = 4$.**Answer: Vertex $(4, 7)$; axis $x = 4$** **Problem 8.2**Solve by factoring: $x^2 - 7x + 12 = 0$ Factor: $(x - 3)(x - 4) = 0$.

Set each factor = 0.

Answer: $x = 3, x = 4$ **Problem 8.3**Solve by quadratic formula: $3x^2 + 5x - 2 = 0$ $a = 3$, $b = 5$, $c = -2$. $D = 5^2 - 4(3)(-2) = 25 + 24 = 49$. $\sqrt{49} = 7$. $x = (-5 \pm 7)/6$. $x = 2/6 = 1/3$, OR $x = -12/6 = -2$.**Answer: $x = 1/3, x = -2$**

Problem 8.4

Find the discriminant of $y = 4x^2 - 12x + 9$ and number of real solutions.

$$D = (-12)^2 - 4(4)(9) = 144 - 144 = 0.$$

Answer: $D = 0$; one (double) real solution.

Problem 8.5

A rocket is launched from a 6-foot-high platform at 64 ft/s. Its height is $h(t) = -16t^2 + 64t + 6$. What is the maximum height?

$$\text{Max at } t = -b/(2a) = -64/(2 \cdot (-16)) = 2 \text{ sec.}$$

$$h(2) = -16(4) + 64(2) + 6 = -64 + 128 + 6 = 70.$$

Answer: Max height = 70 ft

Problem 8.6

Convert to vertex form by completing the square: $y = x^2 - 6x + 11$

$$\text{Group: } y = (x^2 - 6x) + 11.$$

$$\text{Half of } -6 \text{ is } -3; (-3)^2 = 9. \text{ Add and subtract 9 inside: } y = (x^2 - 6x + 9) + 11 - 9.$$

$$y = (x - 3)^2 + 2.$$

Answer: $y = (x - 3)^2 + 2$; vertex (3, 2)

Chapter 9 — Radicals, Pythagorean & Distance

Problem 9.1

Simplify: $\sqrt{72}$

$$\text{Factor out perfect squares: } 72 = 36 \cdot 2.$$

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}.$$

Answer: $6\sqrt{2}$

Problem 9.2

Simplify: $\sqrt{20} + \sqrt{45}$

$$\sqrt{20} = 2\sqrt{5}; \sqrt{45} = 3\sqrt{5}.$$

$$\text{Add like radicals: } 2\sqrt{5} + 3\sqrt{5} = 5\sqrt{5}.$$

Answer: $5\sqrt{5}$

Problem 9.3

Rationalize: $4 / \sqrt{3}$

$$\text{Multiply numerator and denominator by } \sqrt{3}.$$

$$= 4\sqrt{3} / 3.$$

Answer: $4\sqrt{3} / 3$

Problem 9.4

A right triangle has legs 8 and 15. Find the hypotenuse.

$$c^2 = 8^2 + 15^2 = 64 + 225 = 289.$$

$$c = \sqrt{289} = 17.$$

Answer: Hypotenuse = 17

Problem 9.5

Find the distance between (1, -3) and (5, 9).

$$d = \sqrt{[(5-1)^2 + (9-(-3))^2]}$$

$$= \sqrt{[4^2 + 12^2]} = \sqrt{[16 + 144]} = \sqrt{160}.$$

$$\text{Simplify: } \sqrt{160} = \sqrt{(16 \cdot 10)} = 4\sqrt{10}.$$

Answer: $d = 4\sqrt{10}$ (≈ 12.65)

Problem 9.6

Find the midpoint of (-4, 7) and (10, -1).

$$M = ((-4 + 10)/2, (7 + (-1))/2) = (6/2, 6/2) = (3, 3).$$

Answer: Midpoint (3, 3)
